

# Zhaode Wang

AI Inference Engine Expert · On-Device LLM · High-Performance Computing

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## Education

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|------------------------------------------------------------------------------------------------------------------------|-----------------------------------------|-------------|
| <b>Institute of Computing Technology, Chinese Academy of Sciences</b><br>State Key Laboratory of Computer Architecture | M.S. in Computer Architecture           | 2017 - 2020 |
| <b>Shandong University</b>                                                                                             | B.S. in Computer Science and Technology | 2013 - 2017 |

## Experience

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| <b>Alibaba · Taotian Group</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | <b>MNN Tech Lead</b>             | 2020 - Present |
| <ul style="list-style-type: none"><li><b>Inference engine:</b> Lead MNN architecture and performance optimization across CPU, Metal, OpenCL, Vulkan, Hexagon, CoreML, and NNAPI, covering operator kernels, memory planning, graph optimization, and automatic fusion.</li><li><b>On-device LLM:</b> Initiated and lead MNN-LLM for 100+ language and multimodal models, spanning low-bit and mixed-precision quantization, KV-cache and attention optimization, and speculative decoding.</li><li><b>Production impact:</b> Built MNN's on-device Python runtime and enabled dozens of algorithms for Pailitao visual search, Taobao Live, security, and feed; drove MNN deployment across Taobao, Xianyu, Quark, and Qwen, reaching tens of millions of users, and supported DingTalk, Youku, Taobao Instant Commerce, and Amap.</li><li><b>Developer ecosystem:</b> Build end-to-end model conversion and deployment, CV/audio modules, Python APIs, and CI/CD for MNN (15.8K+ GitHub stars).</li></ul> |                                  |                |
| <b>Cambricon Technologies</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | <b>Compiler Engineer Intern</b>  | 2018 - 2020    |
| <ul style="list-style-type: none"><li>Optimized operators for Cambricon accelerators and contributed to compiler optimization and linker development.</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                  |                |
| <b>Megvii</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | <b>Algorithm Engineer Intern</b> | 2019           |
| <ul style="list-style-type: none"><li>Optimized traffic video understanding algorithms and productionized them in C++.</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                  |                |
| <b>Microsoft Asia Engineering Academy</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | <b>Software Engineer Intern</b>  | 2016           |
| <ul style="list-style-type: none"><li>Evaluated LevelDB, Presto, and Hadoop for advertising data storage and analytics workloads.</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                  |                |

## Selected Publications

|                                                                                                                          |         |      |  |
|--------------------------------------------------------------------------------------------------------------------------|---------|------|--|
| <b>Core Contributions</b>                                                                                                |         |      |  |
| Efficient LLM Inference on ARM via Hardware-Aware Operator Co-Design and Heuristic Mixed-Precision Search                | ISCAS   | 2026 |  |
| RecGPT-Mobile: On-Device Large Language Models for User Intent Understanding in Taobao Feed Recommendation               | SIGIR   | 2026 |  |
| MNN-LLM: A Generic Inference Engine for Fast Large Language Model Deployment on Mobile Devices                           | MM Asia | 2024 |  |
| Walle: An End-to-End, General-Purpose, and Large-Scale Production System for Device-Cloud Collaborative Machine Learning | OSDI    | 2022 |  |
| <b>Collaborative Research</b>                                                                                            |         |      |  |
| ARC-Decode: Accelerated Decoding with Risk-Bounded Acceptance                                                            | ICML    | 2026 |  |
| Prism-MoE: Efficient Dense-to-MoE Conversion for Visual Autoregressive Generation                                        | ICML    | 2026 |  |
| AccKV: Towards Efficient Audio-Video LLMs Inference via Adaptive-Focusing and Cross-Calibration KV Cache Optimization    | AAAI    | 2026 |  |
| VertiKV: Vertical-Integrity KV Cache Compression for Efficient Multimodal Long-Context Inference                         | ACM MM  | 2026 |  |
| MadaKV: Adaptive Modality-Perception KV Cache Eviction for Efficient Multimodal Long-Context Inference                   | ACL     | 2025 |  |

## Patents & Applications

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|----------------------------------------------------------------------------------------|-------------------|------|
| BANG-Linker: Heterogeneous Linking Method, Apparatus, and Related Products             | Cambricon · Filed | 2019 |
| BANG Simulator: Programming Debugging Method, Apparatus, and Related Products          |                   |      |
| Solution for Deploying Large Language Models on Mobile Devices                         | Alibaba · Filed   | 2024 |
| Saliency-Aware Blockwise Quantization via Floating-Point Zero-Point Degrees of Freedom | Alibaba · Pending | 2026 |
| Mixed-Precision Allocation for Low-Bit LLMs Based on Action-Conditioned Operator Error |                   |      |

## Honors & Competitions

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|------------------------------------------------------------------------------|--------------------|------------|
| IEEE AICAS Grand Challenge: LLM Software and Hardware System Co-optimization | <b>First Prize</b> | 2024, 2025 |
| NPU Competitions: Tecorigin Operator Development & SOPHGO TPU Programming    | <b>First Prize</b> | 2022, 2024 |

## Skills

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|------------------|---------------------------------------------------------------------------------------------------------------------------|
| <b>Languages</b> | C, C++, Python, Assembly, CUDA, OpenCL, Vulkan, Metal, BANG, Java, Objective-C                                            |
| <b>Expertise</b> | Compilers, AI inference engines, CPU/GPU/NPU operator optimization, model conversion, quantization, large language models |